

WILMINGTON, NC, USA

WMO: 723020

Lat: 34.268N

Lon: 77.900W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13748

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.2	27.5	6.7	7.8	31.8	11.7	10.0	34.7	24.7	53.1	22.1	52.1	5.7	30	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	93.4	77.7	91.2	76.9	89.1	76.3	80.0	88.7	78.9	86.9	78.1	85.5	9.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	144.1	83.4	76.7	139.5	82.6	75.9	135.6	82.0	43.3	88.3	42.3	86.9	41.4	85.7	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.7	16.8	DB	17.7	97.6	3.5	2.1	15.2	99.1	13.1	100.4	11.1	101.6	8.5	103.1	
			WB	15.3	82.2	3.5	1.0	12.8	82.9	10.7	83.4	8.8	84.0	6.3	84.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	46.3	49.1	55.0	63.6	71.0	78.0	80.9	79.6	74.9	65.7	55.8	49.9
	DBStd	14.25	10.34	9.20	8.94	7.61	6.36	4.83	3.70	3.72	4.73	7.38	8.51	9.57
	HDD50	533	197	119	52	3	0	0	0	0	0	1	38	123
	HDD65	2355	581	450	327	117	21	1	0	0	2	83	296	477
	CDD50	5730	84	94	208	411	650	841	957	918	748	489	211	119
	CDD65	2075	3	4	18	75	206	392	492	453	300	106	18	8
	CDH74	17826	7	17	111	485	1513	3553	4958	4220	2177	680	82	23
	CDH80	6014	0	0	13	90	429	1280	1946	1542	602	108	4	0
Wind	WSAvg	7.5	7.9	8.3	8.9	9.2	8.1	7.3	7.1	6.3	7.0	6.6	6.8	7.2
Precipitation	PrecAvg	55.4	4.0	3.7	4.1	3.0	4.5	5.6	7.8	7.2	5.7	2.9	3.1	3.5
	PrecMax	72.0	10.2	11.2	8.3	8.2	8.7	12.9	15.1	14.1	23.4	9.8	7.9	7.0
	PrecMin	37.8	0.7	1.0	0.9	0.5	0.9	0.9	1.7	1.7	0.7	0.4	0.5	0.6
	PrecStd	7.9	2.1	2.0	2.0	1.9	2.2	3.0	3.2	3.6	4.6	2.5	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.9	76.6	81.7	86.8	92.1	95.5	97.0	95.6	91.0	86.4	79.8	77.1
		MCWB	65.5	65.1	65.1	68.2	74.0	76.6	78.5	78.5	75.3	72.4	69.0	67.1
	2%	DB	70.5	72.4	77.1	82.3	87.7	92.3	93.5	92.0	88.2	82.8	76.4	73.0
		MCWB	63.6	62.2	64.0	67.4	72.8	76.5	78.2	78.0	74.6	71.3	66.5	65.9
	5%	DB	67.0	68.6	73.4	79.6	84.6	89.7	91.1	89.7	85.8	80.5	73.3	69.5
		MCWB	61.4	60.4	62.2	65.8	71.5	75.7	77.7	77.2	74.0	70.2	65.3	64.4
	10%	DB	63.5	64.9	69.9	76.4	81.8	87.0	88.8	87.4	83.6	78.0	70.6	66.2
		MCWB	58.5	58.2	61.0	64.5	70.0	74.9	77.1	76.5	73.3	69.5	64.0	61.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.0	68.1	69.4	73.2	76.9	80.3	81.5	81.4	78.7	76.9	72.3	70.8
		MCDB	72.2	73.2	75.7	80.9	86.5	90.7	91.7	90.4	84.8	81.1	75.7	74.1
	2%	WB	65.3	65.6	67.1	70.8	75.0	78.4	80.0	79.9	77.4	75.1	69.9	67.4
		MCDB	69.0	70.3	72.6	77.9	83.7	87.8	89.0	88.0	83.2	79.5	73.8	70.7
	5%	WB	62.5	62.7	65.1	68.9	73.4	77.3	79.0	78.8	76.5	73.2	67.6	65.1
		MCDB	66.2	66.6	70.8	75.5	80.6	85.7	87.3	86.1	82.1	78.0	71.5	68.7
	10%	WB	59.3	59.4	63.0	67.1	72.2	76.4	78.1	77.9	75.5	71.2	65.1	62.3
		MCDB	63.1	64.2	68.6	73.4	79.1	83.8	85.9	84.7	81.0	76.1	69.5	66.0
Mean Daily Temperature Range	5% DB	MDBR	19.7	20.6	21.0	20.5	18.5	16.3	15.6	15.3	15.8	19.1	21.3	19.8
		MCDBR	22.8	23.2	23.4	22.0	20.1	18.6	17.5	17.1	17.7	19.7	22.4	21.2
		MCWBR	16.9	15.1	12.7	10.1	8.2	6.9	6.4	6.4	7.2	9.7	14.1	15.2
	5% WB	MCDBR	20.3	20.9	19.5	18.3	17.1	16.5	16.3	15.5	14.2	14.9	18.5	19.7
		MCWBR	17.0	16.3	13.3	10.4	7.9	7.0	6.9	6.4	6.8	8.7	14.0	16.0
Clear-Sky Solar Irradiance	taub	0.327	0.335	0.366	0.398	0.458	0.488	0.526	0.503	0.427	0.386	0.349	0.331	
	taud	2.517	2.489	2.427	2.324	2.210	2.171	2.079	2.141	2.350	2.435	2.492	2.535	
	Ebn at Noon	280	290	288	282	265	256	246	249	265	268	269	272	
	Edh at Noon	26	30	35	40	46	48	52	48	37	31	27	25	
All-Sky Solar Radiation	RadAvg	856	1080	1431	1800	1972	2038	1930	1738	1460	1241	948	758	
	RadStd	54	102	117	117	136	119	118	104	107	133	70	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.85	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	+265	+189
(47)	Regional (43 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+216

Nomenclature: See separate page